

Daily Practice AP Precalculus Unit 1 and Unit 2

Topics: Rates of Change, Concavity, Sequences, Linear & Exponential Models

Section 0: Core Concepts Review

Unit 1: Rates of Change & Concavity

1. Average Rate of Change (AROC / 平均变化率)

- **【Formula 核心公式】** Over the interval $[a, b]$, the AROC is defined as:

$$AROC = \frac{\Delta y}{\Delta x} = \frac{f(b) - f(a)}{b - a}$$

- **【三大核心意义】**
 1. **Algebraic (代数意义):** Change in y divided by change in x .
 2. **Geometric (几何意义):** The **Slope of the Secant Line (割线的斜率)** connecting points $(a, f(a))$ and $(b, f(b))$.
 3. **Real-world (现实意义):** Average Speed or Average Growth Rate (过程中的平均速度).

2. Concavity (函数的凹凸性)

- **【绝杀法则】** 判断凹凸性, 只看 AROC 的变化趋势 (Look at how AROC changes)!
- **Concave Up (凹向上 U):** AROC is **Increasing** (变化率在不断增大).
- **Concave Down (凹向下 n):** AROC is **Decreasing** (变化率在不断减小).

3. Successive Differences Method (二阶差分法)

- **【适用条件】** Table problems where the x -interval is constant (表格题, x 是等差的).
- **【实战推导】**
 - **1st Diff (一阶差分):** Δy (Consecutive y values subtracted)
 - **2nd Diff (二阶差分):** $\Delta(\Delta y)$ (Difference of the first differences)
 - **If 2nd Diff $> 0 \Rightarrow$ Concave Up U**
 - **If 2nd Diff $< 0 \Rightarrow$ Concave Down n**

4. Linear vs. Quadratic (线性 vs. 二次函数)

- **Linear Functions ($y = mx + b$):** The AROC is **Constant** (一阶差分为常数).
- **Quadratic Functions ($y = ax^2 + bx + c$):** * The AROC itself is a Linear Function! (平均变化率本身是一次函数).
 - The rate of change of the AROC is Constant (二阶差分为常数).

Unit 2: Sequences & Exponential Functions

1. Sequences (数列的区别)

- **Arithmetic (等差):** Constant difference (d). Linear behavior.
- **Geometric (等比):** Constant ratio (r). Exponential behavior.
- **If given non-consecutive terms y_m and y_n (where $m > n$):
 - Arithmetic: $d = \frac{y_m - y_n}{m - n}$ (本质是算两点间的斜率)
 - Geometric: $r = \sqrt[m-n]{\frac{y_m}{y_n}}$ (直接开步数次方的根号)

**2. Exponential Functions:

- **Form:** $f(x) = a \cdot b^x$. (a is initial value, b is base).
- **【底层原理】** 为什么要求 $b > 0$ 且 $b \neq 1$? If $b = 1$, $1^x = 1$, it becomes a flat line. If b is negative (e.g., -2), $(-2)^{1/2} = \sqrt{-2}$ is undefined in real numbers! The graph would break into non-continuous dots.
- **Domain & Range:** Domain is $(-\infty, \infty)$. The Range is restricted by a **Horizontal Asymptote (水平渐近线)**.

3. Growth Rate vs Growth Factor (增长率 vs 增长因子)

- **【易错陷阱】** Value decreases by 12% $\Rightarrow$ You keep $(100\% - 12\%) = 88\%$. The base $b = 0.88$.
- **Exponential Growth:** $b = 1 + r$ ($b > 1$); **Exponential Decay:** $b = 1 - r$ ($0 < b < 1$).

**4. Continuous Compounding

- **【底层推导】** Where does e come from? Imagine 100% interest, compounded infinitely many times a year (每时每刻都在疯狂结息): $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e \approx 2.71828$. It is the absolute limit of continuous growth.
- **Continuous Growth:** Keyword "compounded continuously", directly use the PERT formula: $A = P \cdot e^{rt}$.

5. The AROC of Exponential Functions (指数函数的 AROC 规律)

- While a linear function has a constant AROC, an exponential function's AROC over equal-length intervals grows or decays **proportionally (by the same constant multiple)**. (指数函数在等长区间的AROC也呈等比倍数变化)

Section 1: Multiple Choice Questions

(No calculator is allowed for these questions.)

1. The table below shows values of a function $f(x)$ at constant intervals of x .

x	1	2	3	4
$f(x)$	12	9	5	0

Based on the successive differences, which of the following best describes the concavity of the graph of $f(x)$?

- (A) Concave Up, because the first differences are increasing.
- (B) Concave Up, because the second differences are positive.
- (C) Concave Down, because the first differences are decreasing.
- (D) Concave Down, because the second differences are negative.

2. Which of the following is the best geometric interpretation of the Average Rate of Change of a function $f(x)$ over the interval $[a, b]$?

- (A) The slope of the tangent line at $x = a$.
- (B) The y-intercept of the secant line connecting $(a, f(a))$ and $(b, f(b))$.
- (C) The slope of the secant line connecting $(a, f(a))$ and $(b, f(b))$.
- (D) The maximum value of $f(x)$ between $x = a$ and $x = b$.

3. A function $f(x)$ has an average rate of change of 4 over the interval $[0, 2]$, an average rate of change of 8 over the interval $[2, 4]$, and an average rate of change of 12 over the interval $[4, 6]$. Which type of function best models $f(x)$?

- (A) Linear
- (B) Quadratic
- (C) Exponential Growth
- (D) Exponential Decay

4. A sequence has $g_3 = 24$ and $g_6 = 192$. If the sequence is geometric and the common ratio $r > 0$, what is the value of g_8 ?

- (A) 384
- (B) 576
- (C) 768
- (D) 1536

5. A sequence has $a_3 = 10$ and $a_8 = 25$. If the sequence is arithmetic, what is the value of a_{10} ?

- (A) 28
- (B) 31

(C) 35

(D) 40

6. A function $f(x)$ has the following values given in the table below:

x	1	4	7
$f(x)$	5	20	80

Which of the following statements best justifies why $f(x)$ is an exponential function?

(A) Because equal length input intervals yield equal length output differences.

(B) Because equal length input intervals yield proportional output values.

(C) Because the function has a constant rate of change.

(D) Because the second differences of the outputs are constant.

7. The population of a city decreases by exactly 4% each year. If the current population is P_0 , which of the following functions $P(t)$ best models the population t years from now?

(A) $P(t) = P_0(0.04)^t$

(B) $P(t) = P_0(0.96)^t$

(C) $P(t) = P_0(-0.04)^t$

(D) $P(t) = P_0(1.04)^{-t}$

8. Which of the following functions is equivalent to $h(x) = 2 \cdot 5^{x-1}$?

(A) $h(x) = 10^{x-1}$

(B) $h(x) = \frac{2}{5} \cdot 5^x$

(C) $h(x) = 10 \cdot 5^x$

(D) $h(x) = -3 \cdot 5^x$

9. Consider the function $f(x) = -3 + 4(0.2)^x$. What is $\lim_{x \rightarrow \infty} f(x)$?

(A) $-\infty$

(B) -3

(C) 0

(D) 1

10. The graph of the exponential function $f(x) = 3^x$ is horizontally translated to the right by 2 units to create a new function $g(x)$. Which of the following shows $g(x)$ expressed as a vertical dilation of $f(x)$?

(A) $g(x) = 9 \cdot 3^x$

(B) $g(x) = 6 \cdot 3^x$

(C) $g(x) = \frac{1}{6} \cdot 3^x$

(D) $g(x) = \frac{1}{9} \cdot 3^x$

11. The half-life of a radioactive isotope is 12 days. If the initial mass is 100 grams, which of the following expressions represents the mass remaining after 36 days?

(A) $100\left(\frac{1}{2}\right)^{432}$

(B) $100\left(\frac{1}{2}\right)^{36}$

(C) $100\left(\frac{1}{2}\right)^3$

(D) $100\left(\frac{1}{2}\right)^{\frac{1}{3}}$

12. An exponential function $f(x)$ has the property that its average rate of change over the interval $x = 0$ to $x = 2$ is 15. Its average rate of change over the interval $x = 2$ to $x = 4$ is 60. What is the average rate of change of $f(x)$ over the interval $x = 4$ to $x = 6$?

(A) 105

(B) 120

(C) 240

(D) 900

13. An exponential function is given by $f(x) = a \cdot b^x$. If $f(2) = 20$ and $f(5) = 160$, what is the value of the initial amount a ?

(A) 2

(B) 4

(C) 5

(D) 10

14. An investment grows continuously at an annual rate r . If the initial investment triples in exactly t years, which of the following equations must be true?

(A) $3 = e^{rt}$

(B) $3 = 1 + e^{rt}$

(C) $t = 3e^r$

(D) $e^3 = rt$

Section 2: Free Response Questions

(Show all your work clearly.)

FRQ 1

A researcher is studying the temperature $T(h)$ of a chemical reaction over time h , measured in hours. The data collected is shown in the table below.

h (hours)	0	2	4	6
$T(h)$ ($^{\circ}\text{C}$)	10	16	26	40

(a) Calculate the Average Rate of Change (AROC) of the temperature over the interval $[0, 2]$, the interval $[2, 4]$, and the interval $[4, 6]$. Include appropriate units in your answers.

(b) Based on your calculations in part (a), determine if the graph of $T(h)$ is concave up or concave down. Justify your reasoning using the behavior of the AROC.

(c) The researcher suspects that $T(h)$ can be modeled exactly by a quadratic function. By calculating the second differences of the table, prove whether the researcher's suspicion is correct.

FRQ 2

Let $M(t)$ be the mass of a radioactive substance, in grams, remaining after t days. At $t = 0$, the mass is 400 grams. At $t = 3$ days, the mass is 50 grams.

(a) Assuming $M(t)$ is modeled by the discrete exponential form $M(t) = a \cdot b^t$, find the exact values of the constants a and b .

(b) A scientist proposes an alternative model using continuous decay: $N(t) = 400 \cdot e^{-kt}$. By comparing $M(t)$ and $N(t)$, find the exact value of k . (Leave your answer in terms of the natural logarithm $\ln$).

(c) Using the model $M(t)$ found in part (a), determine the exact time t (in days) when the mass of the substance reaches 25 grams. Show your reasoning algebraically.

FRQ 3

Consider the function $f(x) = 12 \cdot 2^{2x-3} - 5$.

- (a)** Rewrite the exponential part of $f(x)$ to express the entire function in the standard form $f(x) = a \cdot b^x + c$, where a , b , and c are constants. Show all steps of your algebraic manipulation.
- (b)** Based on your rewritten function from part (a), determine the domain and range of $f(x)$, and state the equation of its horizontal asymptote.
- (c)** Set up and solve the equation $f(x) = 19$ to find the exact value of x . Show all algebraic steps.

FRQ 4

The number of users on a new social media platform is recorded over time. At year $t = 0$, there are 200 users. At year $t = 2$, there are 800 users. Two data analysts are proposing different models for the growth of the users.

- (a)** Analyst A proposes a linear model, $L(t)$. Find the explicit equation for $L(t)$ that models the number of users t years after launch.
- (b)** Analyst B proposes an exponential model, $E(t) = a \cdot b^t$. Find the exact values of a and b to complete the model.
- (c)** Using the exponential model $E(t)$ found in part (b), what is the annual percent rate of increase for the platform's users? Explain how you found this from your base b .
- (d)** Calculate the Average Rate of Change (AROC) of the exponential model $E(t)$ over the interval from $t = 0$ to $t = 2$. Then, calculate the AROC of $E(t)$ over the next equal-length interval from $t = 2$ to $t = 4$. Compare these two values and explain what this demonstrates about the rate of change of an exponential function.

FRQ 5

Two sequences are given: an arithmetic sequence a_n and a geometric sequence g_n . For the arithmetic sequence, $a_2 = 7$ and $a_5 = 16$. For the geometric sequence, $g_2 = 6$ and $g_4 = 54$ (with $r > 0$).

- (a)** Find the common difference d for sequence a_n , and write the explicit formula for a_n .
- (b)** Find the common ratio r for sequence g_n , and write the explicit formula for g_n .
- (c)** Suppose a continuous exponential function $f(x) = A \cdot B^x$ is created such that $f(2) = g_2$ and $f(4) = g_4$. Evaluate the ratio $\frac{f(3.5)}{f(2.5)}$ and explain the mathematical meaning of this ratio in relation to the base B .